Problem 4.13

A fund is earning 5% simple interest. The amount in the fund at the end of the 5th year is \$10,000. Calculate the amount in the fund at the end of 7 years.

Solution

Since the fund earns simple interest,

$$A(t) = P(1 + it), \qquad i = 0.05.$$

At the end of 5 years, the amount is \$10,000, so

$$10{,}000 = P(1 + 0.05 \cdot 5) = P(1.25).$$

Hence,

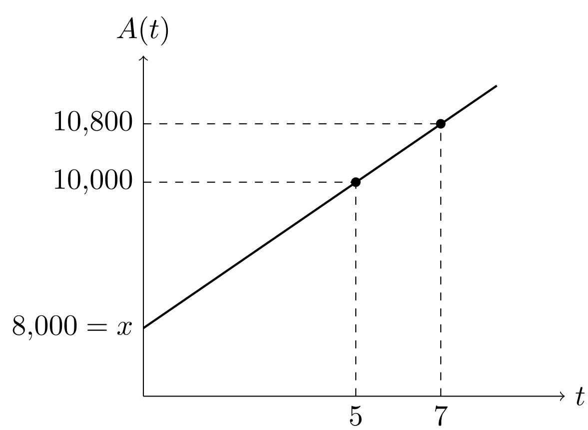
$$P = \frac{10{,}000}{1.25} = 8{,}000.$$

Now evaluate the amount at the end of 7 years:

$$A(7) = 8{,}000(1 + 0.05 \cdot 7) = 8{,}000(1.35) = 10{,}800.$$

Therefore, the amount in the fund at the end of 7 years is

$$\boxed{\$10{,}800}.$$



$A(t) = 8000(1 + (0.05)t)$